

PROJECT REPORT OF CO-OP Project at Industry (Module-2)
ON
A Hybrid AI-Enhanced Transaction Monitoring Framework for Fraud Detection
submitted in partial fulfilment of the requirements for the award of degree of
BACHELOR OF ENGINEERING
In
COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled “**A Hybrid AI-Enhanced Transaction Monitoring Framework for Fraud Detection**” in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Isha Kansal**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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ACKNOWLEDGEMENT

We would like to express our sincere gratitude to our mentor and faculty members for their continuous guidance, encouragement, and support throughout the completion of this project. Their valuable suggestions and technical expertise helped us understand the practical implementation of Artificial Intelligence and Fraud Detection Systems in modern financial ecosystems.

We are also thankful to Chitkara University for providing us with the opportunity to work on this industry-oriented research project. This project significantly enhanced our knowledge in areas such as Machine Learning, Cybersecurity, Data Analytics, and Financial Fraud Detection.

We would also like to thank our peers and teammates for their cooperation, discussions, and contributions throughout the development and research process.

ABSTRACT

The rapid expansion of digital payment systems, online banking platforms, and financial technologies has significantly transformed the global financial ecosystem. Technologies such as

Unified Payments Interface (UPI), online wallets, debit cards, credit cards, and mobile banking have made digital transactions more convenient, efficient, and accessible for users worldwide. However, this rapid digital transformation has also increased the risks of cyber fraud, unauthorized transactions, phishing attacks, fake QR scams, SIM swapping, account hacking, and identity theft.

Traditional fraud detection systems mainly depend upon rule-based transaction monitoring techniques that often fail to identify modern sophisticated fraud patterns. These systems generally generate high false positives and lack the capability to perform behavioral and network-level fraud analysis in real time.

This project proposes a **Hybrid AI-Enhanced Transaction Monitoring Framework for Fraud Detection** that integrates:

- Rule-Based Detection
- Machine Learning Algorithms
- Anomaly Detection
- Graph-Based Fraud Analysis

The proposed framework evaluates transaction patterns, user behavior, device information, IP addresses, transaction frequency, location patterns, and interconnected account relationships to identify suspicious financial activities with high accuracy and reduced false alarms.

The experimental results demonstrated that the hybrid framework achieved superior fraud detection performance compared to traditional systems. The framework successfully minimized false positives while improving fraud detection capability in real-time financial systems.

The project highlights the importance of integrating Artificial Intelligence into financial cybersecurity systems to ensure secure, scalable, and efficient digital financial transactions.

1. INTRODUCTION

The rapid growth of digital technologies has transformed the financial sector worldwide. In India, the increasing use of digital payment systems such as Net Banking, Unified Payments Interface

(UPI), credit and debit cards, mobile wallets, and online banking platforms has improved the speed, accessibility, and convenience of financial transactions. Government initiatives promoting cashless transactions have further accelerated the adoption of digital payments across urban and rural regions.

However, the growth of digital financial systems has also increased cybersecurity threats and financial frauds. Cybercriminals exploit system vulnerabilities using techniques such as phishing, QR hijacking, SIM swapping, malware attacks, social engineering, and identity theft. Since most transactions occur in real time, detecting and preventing fraudulent activities has become extremely challenging.

Traditional fraud detection systems mainly rely on predefined rule-based monitoring methods such as unusual transaction amounts, suspicious login behavior, or high transaction frequency.

Machine Learning techniques have improved fraud detection by analyzing transaction behavior and identifying hidden suspicious patterns. However, standalone machine learning models still face challenges related to transparency, explainability, computational complexity, and real-time implementation.

Modern financial fraud often involves interconnected transaction networks where individual transactions may appear normal but collectively indicate organized fraudulent activity. Therefore, there is a growing need for an advanced Hybrid AI-based Fraud Detection System capable of realtime monitoring, behavioral analysis, network-level fraud detection, and reduced false positives.

This project proposes a **Hybrid AI-Enhanced Transaction Monitoring Framework** designed to improve fraud detection accuracy and strengthen cybersecurity in modern digital financial systems.

2. LITERATURE REVIEW

Numerous research studies have explored fraud detection systems using Artificial Intelligence, Machine Learning, and Deep Learning techniques.

2.1 AI-Based Fraud Detection Methods

Early fraud detection systems primarily used rule-based approaches that relied on predefined thresholds to detect suspicious activities. However, these systems lacked flexibility and adaptability.

Bhattacharyya et al. applied:

- Logistic Regression
- Random Forest
- Support Vector Machines (SVM)

for credit card fraud detection. Their research showed that Random Forest achieved better fraud detection accuracy compared to other models.

Dal Pozzolo et al. addressed the issue of imbalanced fraud datasets using ensemble learning techniques and preprocessing methods.

Whitrow et al. emphasized transaction aggregation and behavioral feature engineering for improving fraud detection performance.

2.2 Deep Learning-Based Fraud Detection

Recent advancements in Deep Learning introduced advanced models such as:

- Recurrent Neural Networks (RNN)
- Convolutional Neural Networks (CNN)
- Long Short-Term Memory (LSTM)

These models can capture complex sequential transaction patterns and identify suspicious behavior more accurately.

Hybrid AI models combining multiple techniques were also proposed to improve detection efficiency.

2.3 Graph-Based Fraud Detection

Modern fraud activities often involve interconnected account networks.

Graph-based fraud detection techniques model:

- Nodes = Accounts
- Edges = Transactions

This helps identify:

- Mule account chains
- Fraud communities
- Suspicious transaction networks

Graph mining techniques significantly improve organized fraud detection capability.

3. MATERIALS AND METHODOLOGY

3.1 Dataset

The dataset used in this project contains both legitimate and fraudulent financial transaction records.

Features included:

Feature Name	Description
Transaction ID	Unique transaction identifier
Transaction Amount	Amount transferred
Transaction Time	Time of transaction
Device ID	Device information
IP Address	User IP address
Location	Geographical location
Transaction Type	UPI/Card/Wallet
Beneficiary ID	Receiver identifier
Transaction Count	Number of transactions
Target	Fraud or Normal

The dataset reflects real-world financial transaction characteristics.

3.2 Data Preprocessing

Data preprocessing was performed to improve model performance and accuracy.

Steps included:

1. Removing Data Leakage

Unique identifiers causing overfitting were removed.

2. Handling Missing Values

Missing data was handled using:

- Mean Imputation
- Median Imputation
- Mode Imputation

3. Feature Encoding

Categorical values were converted using:

- Label Encoding
- One-Hot Encoding

4. Feature Scaling

Standardization techniques normalized feature ranges.

5. Train-Test Split

Dataset division:

- 80% Training Data
- 20% Testing Data

4. HYBRID AI FRAMEWORK

The proposed hybrid framework integrates multiple fraud detection techniques:

Components:

- Rule-Based Engine
- Machine Learning Models
- Anomaly Detection
- Graph-Based Fraud Analysis

Rule-Based Detection

Ensures regulatory compliance and predefined fraud checks.

Machine Learning

Detects hidden behavioral fraud patterns.

Anomaly Detection

Identifies unusual transaction behavior.

Graph Analysis

Detects organized fraud networks and interconnected suspicious accounts.

5. MACHINE LEARNING ALGORITHMS

5.1 Logistic Regression

A supervised machine learning algorithm used for binary classification problems.

Advantages:

- Simple
- Fast
- Interpretable
- Limitations:
 - Poor performance on complex fraud patterns

5.2 Random Forest

An ensemble learning algorithm consisting of multiple decision trees.

Advantages:

- Better accuracy
- Reduced overfitting
- Robust performance

5.3 XGBoost

A gradient boosting algorithm designed for high performance.

Advantages:

- High detection accuracy
- Better optimization
- Handles large datasets efficiently

6. EXPERIMENTAL SETUP

Tools Used

Tool	Purpose
Python 3.x	Programming
Pandas	Data Handling
NumPy	Numerical Operations
Scikit-learn	Machine Learning
XGBoost	Gradient Boosting
Matplotlib	Visualization
Seaborn	Statistical Graphs

Steps Performed

1. Data Loading
2. Data Preprocessing
3. Feature Engineering
4. Model Training
5. Fraud Detection
6. Performance Evaluation
7. Visualization

7. RESULTS AND DISCUSSION

Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	94%	93%	92%	92%
Random Forest	98%	97%	97%	97%
XGBoost	99%	99%	98%	98%
Hybrid Framework	99.5%	99%	99%	99%

Analysis

Logistic Regression

Performed well but struggled with complex non-linear fraud patterns.

Random Forest

Improved fraud detection using feature interactions.

XGBoost

Achieved excellent fraud detection accuracy with optimized performance.

Hybrid Framework

Delivered the best results by integrating:

- Rule-Based Detection
- Machine Learning
- Anomaly Detection
- Graph Analysis

The hybrid model significantly reduced false positives and improved fraud detection efficiency.

8. MAJOR FINDINGS

1. Behavioral transaction patterns are highly useful in fraud detection.
2. Ensemble learning models outperform traditional linear models.
3. Hybrid AI systems reduce false positives significantly.
4. Graph-based analysis effectively identifies organized fraud networks.
5. Real-time fraud detection improves financial cybersecurity.
6. XGBoost and Random Forest showed superior fraud detection capability.

9. LIMITATIONS

Despite strong performance, the project has certain limitations:

- Dependency on dataset quality
- Binary classification limitation
- Computational complexity
- Limited real-time deployment testing
- High resource requirements for graph analysis

10. CONCLUSION AND FUTURE SCOPE

This project proposed a Hybrid AI-Enhanced Transaction Monitoring Framework for detecting fraud in digital financial systems.

The proposed framework successfully combined:

- Rule-Based Systems
- Machine Learning
- Anomaly Detection

- Graph-Based Analysis

The hybrid approach demonstrated:

- High fraud detection accuracy
- Reduced false positives
- Better scalability
- Improved real-time monitoring capability

The framework provides an effective and sustainable solution for modern digital financial ecosystems.

Future Scope

Future improvements may include:

- Real-time banking deployment • Blockchain integration
- Multi-class fraud classification
- Deep Learning optimization
- Graph Neural Networks
- Faster lightweight fraud detection architectures

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